

DC-URS v1 — source of truth

Status: CEO-approved specification for the Delta Climate Urban Resilience Score v1 engine. **Provenance:** authored by the CEO, captured 2026-07-27 from the shared Gemini conversation

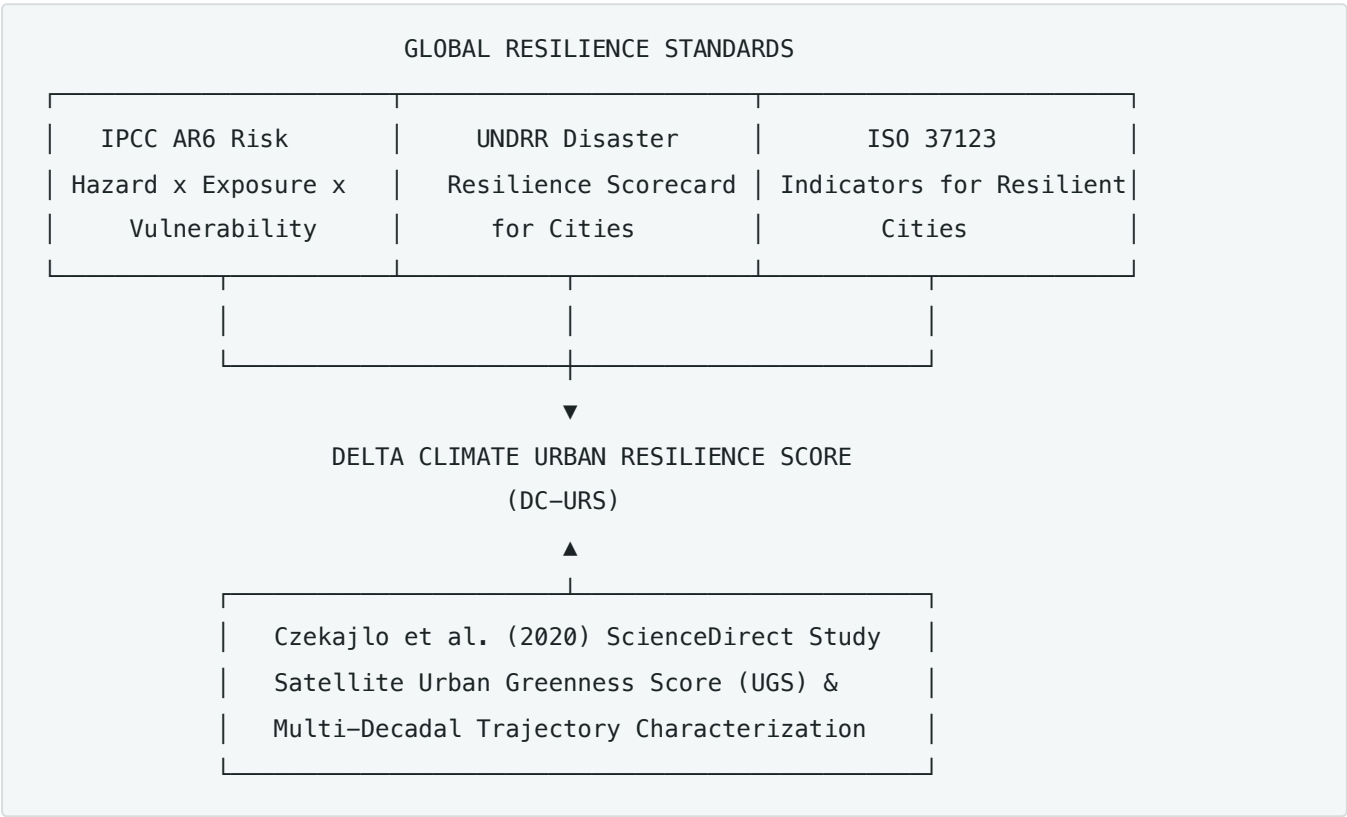
gemini.google.com/share/7fd738ca3e98. **PRECEDENCE (decided 2026-07-27): this document is the FINAL source of truth for v1.** The engine ships it as written, plus exactly three fixes — each a case where this document contradicts itself or the maths misbehaves. The other five findings in [dc-urs-engineering-review.md](#) are deferred to v2.

Formulas below were de-mangled from the source page's rendering. Content is otherwise verbatim.

1. Executive summary and theoretical foundations

The **Delta Climate Urban Resilience Score (DC-URS)** is an integrated composite index designed to assess, benchmark, and simulate urban climate resilience at hyper-local spatial scales (pixel/block level up to municipal ward boundaries).

The methodology synthesises multi-decadal satellite observation techniques with global city resilience benchmarks:



Core benchmark syntheses

1. **Czekajlo et al. (2020) — Urban Greenness Score (UGS)**: incorporates multi-decadal Landsat/Sentinel vegetative trajectory analysis (NDVI persistence, canopy density, and greenness stability) as a fundamental driver of eco-physiological cooling capacity.
2. **IPCC AR6 Risk Framework**: structures the scoring function around the triad of **Hazard (H)**, **Exposure (E)**, and **Vulnerability (V)**, where $V = f(\text{Sensitivity } S, \text{Adaptive Capacity } A)$.
3. **UNDRR & ISO 37123 / ISO 37120**: standardised operational indicators for built-environment durability, urban canopy coverage, heat-refuge accessibility, and critical infrastructure stress.
4. **C40 Cities Heat Resilient City Framework**: informs target thresholds for surface albedo, nature-based cooling solutions, and vulnerable population protection.

2. Mathematical formulation and architecture

DC-URS is computed on a scale of 0 to 100, where 100 represents maximum climate resilience and 0 indicates extreme vulnerability to thermal and climate shocks.

$$\text{DC-URS} = 100 \times [w_A \cdot \text{ACI} + w_E \cdot (1 - \text{EVI}) + w_H \cdot (1 - \text{THI})]$$

Where:

- $\text{ACI} \in [0, 1]$ — **Adaptive Capacity Index** (positive resilience pillar)
- $\text{EVI} \in [0, 1]$ — **Exposure & Sensitivity Index** (negative impact pillar)
- $\text{THI} \in [0, 1]$ — **Thermal Hazard Index** (negative hazard pillar)
- w_A, w_E, w_H — weighting coefficients, $\sum w_i = 1.0$. Default calibration: $w_A = 0.40$, $w_E = 0.35$, $w_H = 0.25$

3. Detailed sub-index formulations

Pillar 1 — Thermal Hazard Index (THI)

Quantifies localised micro-climatic intensity derived from satellite thermal channels and meteorological forcing.

$$\text{THI} = \omega_1 \cdot \text{LST}_{\text{day}} + \omega_2 \cdot \text{LST}_{\text{night}} + \omega_3 \cdot \text{UHI}_{\Delta}$$

- LST_{day} (daytime land surface temperature): peak Landsat 8/9 TIRS Band 10 thermal readings.
- $\text{LST}_{\text{night}}$ (nighttime heat retention): Sentinel-3 SLSTR or MODIS night surface temperature (captures thermal mass trapping).

- **UHI_Δ** (urban heat island delta): localised surface temperature excess relative to a rural baseline T_{rural} :

$$\text{UHI}_\Delta = \text{LST}_{\text{pixel}} - T_{\text{rural}}$$

Note: all variables X are normalised using min-max scaling:

$$X_{\text{norm}} = (X - X_{\text{min}}) / (X_{\text{max}} - X_{\text{min}})$$

Pillar 2 — Exposure & Sensitivity Index (EVI)

Measures spatial vulnerability of population and built assets.

$$\text{EVI} = \beta_1 \cdot \rho_{\text{pop}} + \beta_2 \cdot \text{FAR} + \beta_3 \cdot \text{HVI}_{\text{socio}}$$

- ρ_{pop} (population density): inhabitants per km².
- **FAR** (floor-area ratio / urban density): total built floor area divided by land lot area.
- **HVI_socio** (socio-economic heat vulnerability index): composite ratio of elderly (>65), children (<5), low-income demographics, and informal settlement fraction.

Pillar 3 — Adaptive Capacity Index (ACI), incorporating satellite UGS

The cornerstone pillar, integrating Czekajlo et al.'s **Urban Greenness Score** alongside structural retrofits.

$$\text{ACI} = \alpha_1 \cdot \text{UGS} + \alpha_2 \cdot \text{CRI} + \alpha_3 \cdot \text{TRA}$$

A. Satellite Urban Greenness Score (UGS)

Derived from multi-decadal time-series analysis (Czekajlo et al., 2020) to capture vegetation health, stability, and canopy coverage:

$$\text{UGS} = \varphi_1 \cdot \text{NDVI} + \varphi_2 \cdot \text{FVC} + \varphi_3 \cdot \text{VSI}$$

1. Mean vegetation trajectory (NDVI):

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

2. Fractional Vegetation Cover (FVC):

$$\text{FVC} = (\text{NDVI} - \text{NDVI}_{\text{bare}}) / (\text{NDVI}_{\text{veg}} - \text{NDVI}_{\text{bare}})$$

3. **Vegetation Stability Index (VSI)**: measures long-term vegetative persistence versus land degradation across a multi-year baseline:

$$VSI = 1 - (\sigma_{NDVI} / NDVI)$$

B. Cool Albedo & Structural Reflectivity Index (CRI)

Quantifies surface albedo from satellite shortwave infrared (SWIR) and visible bands to capture cool roof retrofits:

$$\text{Albedo} = 0.356 \cdot \alpha_{\text{Blue}} + 0.130 \cdot \alpha_{\text{Red}} + 0.373 \cdot \alpha_{\text{NIR}} + 0.085 \cdot \alpha_{\text{SWIR1}} + 0.055 \cdot \alpha_{\text{SWIR2}}$$

C. Thermal Refuge Accessibility (TRA)

Proximity to public cooling centres, shaded parks, and urban water bodies (Hooghly River, wetlands):

$$TRA = \exp(-\lambda \cdot d_{\text{cool}})$$

Where d_{cool} is distance in metres to the nearest thermal refuge, $\lambda \approx 0.002$.

4. Indicator weighting matrix

Weights can be statically assigned via Analytic Hierarchy Process (AHP) or calculated dynamically using the Entropy Weighting Method (EWM):

Index pillar	Sub-indicator	Symbol	Default weight w_i	Standard alignment
Hazard (THI)	Daytime surface temp	LST_day	0.10	IPCC AR6 / C40
	Nighttime retained heat	LST_night	0.10	IPCC AR6 / C40
	UHI anomaly delta	UHI_Δ	0.05	ISO 37123
Exposure (EVI)	Population density	ρ_{pop}	0.15	UNDRR Scorecard
	Built volume (FAR)	FAR	0.10	ISO 37120
	Socio-demographic vulnerability	HVI_socio	0.10	IPCC AR6 / ISO 37123
Adaptive capacity (ACI)	Urban Greenness Score	UGS	0.20	Czekajlo et al. (2020)
	Surface albedo index	CRI	0.10	C40 / ISO 37123
	Thermal refuge accessibility	TRA	0.10	UNDRR Scorecard

5. Reference implementation

```
import numpy as np
import pandas as pd

class DeltaClimateResilienceEngine:
    """
    Holistic Urban Resilience Calculator based on Czekajlo et al. (2020)
    and IPCC AR6 / UNDRR / ISO 37123 Standards.
    """
    def __init__(self, weights=None):
        self.weights = weights or {
            'w_A': 0.40, 'w_E': 0.35, 'w_H': 0.25,
            'phi_1': 0.40, 'phi_2': 0.40, 'phi_3': 0.20, # UGS weights
        }

    def calculate_ugs(self, ndvi_mean, fvc, ndvi_std):
        """Calculates Urban Greenness Score (Czekajlo et al. 2020)"""
        # Vegetation Stability Index (VSI)
        vsi = 1.0 - np.clip(ndvi_std / (ndvi_mean + 1e-6), 0.0, 1.0)

        ugs = (
            self.weights['phi_1'] * np.clip(ndvi_mean, 0, 1) +
            self.weights['phi_2'] * np.clip(fvc, 0, 1) +
            self.weights['phi_3'] * vsi
        )
        return ugs

    def calculate_resilience_score(self, lst_day, lst_night, rural_base,
                                   pop_density, far, socio_vulnerability,
                                   ndvi_mean, fvc, ndvi_std, albedo, dist_cool_refuge):
        """
        Computes composite DC-URS Score (0 - 100)
        """
        # 1. Thermal Hazard Index (THI)
        uhi_delta = max(0.0, lst_day - rural_base)
        thi = (
            0.40 * np.clip((lst_day - 25.0) / 20.0, 0, 1) +
            0.40 * np.clip((lst_night - 20.0) / 15.0, 0, 1) +
            0.20 * np.clip(uhi_delta / 10.0, 0, 1)
        )
```

```

# 2. Exposure & Vulnerability Index (EVI)
evi = (
    0.45 * np.clip(pop_density / 25000.0, 0, 1) +
    0.30 * np.clip(far / 5.0, 0, 1) +
    0.25 * np.clip(socio_vulnerability / 10.0, 0, 1)
)

# 3. Adaptive Capacity Index (ACI)
ugs = self.calculate_ugs(ndvi_mean, fvc, ndvi_std)
cri = np.clip(albedo / 0.60, 0, 1)
tra = np.exp(-0.002 * dist_cool_refuge)

aci = (0.50 * ugs) + (0.30 * cri) + (0.20 * tra)

# Final Composite Calculation
dc_urs = 100.0 * (
    self.weights['w_A'] * aci +
    self.weights['w_E'] * (1.0 - evi) +
    self.weights['w_H'] * (1.0 - thi)
)

return {
    'DC_URS': round(float(dc_urs), 2),
    'ACI': round(float(aci), 3),
    'EVI': round(float(evi), 3),
    'THI': round(float(thi), 3),
    'UGS': round(float(ugs), 3)
}

# Example usage for Ballygunge vs Baruipur
engine = DeltaClimateResilienceEngine()

# Ballygunge (Dense urban core)
ballygunge = engine.calculate_resilience_score(
    lst_day=42.5, lst_night=31.0, rural_base=32.0,
    pop_density=22000, far=3.8, socio_vulnerability=7.2,
    ndvi_mean=0.18, fvc=0.12, ndvi_std=0.08, albedo=0.15, dist_cool_refuge=800
)

# Baruipur (Peri-urban green space)
baruipur = engine.calculate_resilience_score(
    lst_day=34.1, lst_night=24.5, rural_base=32.0,
    pop_density=4500, far=0.8, socio_vulnerability=4.1,

```

```
ndvi_mean=0.58, fvc=0.54, ndvi_std=0.04, albedo=0.22, dist_cool_refuge=250
)

print("Ballygunge Resilience Score:", ballygunge)
print("Baruipur Resilience Score:", baruipur)
```

Verified output (run 2026-07-27):

```
Ballygunge {'DC_URS': 20.01, 'ACI': 0.231, 'EVI': 0.804, 'THI': 0.843, 'UGS': 0.231}
Baruipur   {'DC_URS': 65.23, 'ACI': 0.548, 'EVI': 0.231, 'THI': 0.344, 'UGS': 0.634}
```

6. Tiered classification system for UI visualisation

Score range	Resilience tier	UI colour	Actionable guidance
80.0 – 100.0	Optimal Resilience	#22c55e green	Maintain canopy cover and protect ecosystem services.
60.0 – 79.9	Moderate Resilience	#eab308 yellow	Expand cool roofs and targeted tree corridors.
40.0 – 59.9	Vulnerable	#f97316 orange	Urgent intervention required: pocket parks & high-albedo coatings.
0.0 – 39.9	Critical Hotspot	#ef4444 red	Extreme emergency risk: deploy urban greening and cooling centres.

What v1 ships

IN — three fixes. Each is a case where this document contradicts itself or the maths misbehaves, so v1 cannot ship without them.

#	Fix	Why it is not optional
3	VSI guarded at NDVI_mean > 0.10	Unguarded it returns 1.00 for water: negative NDVI → negative ratio → clip floors it at 0 → 1 - 0 = 1 . The Hooghly would score as perfect vegetation stability
5	§5 code wins over the §4 table on pillar weights	The two disagree on EVI and ACI. The code is what produced this document's published 20.01 / 65.23
6	Fixed anchors, not min-max	§3 says min-max, §5 uses fixed divisors. Min-max would silently rescore every existing ward when a fourth is added

OUT — deferred to v2 by decision. Each is marked *KNOWN LIMITATION* for v2 at the point in `dc-urs.ts` where it bites, and pinned by an assertion so it cannot be changed silently.

#	Deferred	Effect while deferred
1	Geometric aggregation	The weighted sum permits full compensation : a ward at THI 0.85 / ACI 0.90 scores 8.3 points above one at THI 0.30 / ACI 0.35
2	Drop the redundant FVC term	FVC is an affine transform of NDVI by §3's own definition, so 0.80 of the greenness weight rests on one variable counted twice
4	Canopy fraction as an independent input	Collected in Phase 1 but inert in v1; an assertion pins it inert
7	/25 daytime anchor	The /20 term saturates at 45 °C, so the index cannot discriminate above that
8	Czekajlo attribution wording	Documentation only; no code effect

Verified: the shipped engine reproduces this document's own worked examples exactly — **Ballygunge 20.01**, **Baruipur 65.23**.

Settled: three wards for v1; WorldPop + NFHS-5/Census 2011 for demographics; anchors as specified here; tier labels ship as written.